

AI CAREER FOR WOMEN

Engineer Spoke



AI DSA TUTOR

A Project Report
submitted in partial fulfillment of the requirements
of
AICW project.

by

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ABSTRACT of the Project

Data Structures and Algorithms (DSA) are fundamental components of computer science and are essential for programming, problem-solving, competitive programming, and technical interviews. However, students often find it difficult to select problems according to their skill level and may require guidance while solving challenging problems. Existing practice platforms generally provide problems and solutions, but they may not offer a structured learning experience with difficulty prediction and progressive assistance.

To address this problem, **AI DSA Tutor** is developed as an interactive web-based DSA learning platform that combines DSA practice with Machine Learning. The system provides a collection of **30 DSA problems covering 11 topics**, categorized into Easy, Medium, and Hard levels. A **Decision Tree-based Machine Learning model** is incorporated to predict the difficulty level of DSA problems using selected features. The system also provides a confidence score associated with the prediction.

The platform includes **progressive hints**, allowing students to receive guidance step-by-step rather than immediately viewing a complete solution. It also provides example inputs and outputs along with **time and space complexity analysis**, helping students understand both problem-solving logic and algorithm efficiency.

The application is developed using **Python, Pandas, Scikit-learn, and Streamlit**. The final system provides an interactive and user-friendly environment for practicing DSA and demonstrates how Machine Learning can be applied to support educational and problem-solving applications.

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CHAPTER 1

Introduction

3.2 Problem Statement

Data Structures and Algorithms are essential subjects in Computer Science and are widely used in software development, competitive programming, and technical interviews. Although students can access a large number of DSA problems through online platforms, beginners often face difficulty in identifying problems according to their current skill level and understanding the correct approach to solve them.

A common challenge is that students may either receive too little guidance or directly look at complete solutions without attempting the problem themselves. This can reduce their opportunity to develop independent problem-solving skills. Students also need to understand the efficiency of an algorithm in terms of time and space complexity.

Therefore, there is a need for a learning platform that combines DSA practice with guided assistance. The proposed AI DSA Tutor addresses this problem by providing categorized DSA problems, Machine Learning-based difficulty prediction, progressive hints, examples, and complexity analysis through a single interactive platform.

1.2 Motivation

The motivation behind developing AI DSA Tutor is to make DSA learning more structured and interactive for students.

Many students practice DSA for academic examinations, coding competitions, placements, and technical interviews. However, selecting suitable problems and understanding how much help is required can be challenging.

The project aims to provide a learning environment where students can:

- Select DSA topics according to their interests.
- Practice problems at different difficulty levels.

- Attempt problems independently.
- Reveal hints gradually when they need assistance.
- Understand time and space complexity.
- Obtain an AI-based prediction of problem difficulty.

The integration of Machine Learning into the platform also demonstrates a practical application of classification techniques in the field of education.

1.3 Objectives

The major objectives of the project are:

1. To develop an interactive web-based platform for DSA practice.
 2. To provide DSA problems from multiple important topics.
 3. To categorize problems into Easy, Medium, and Hard difficulty levels.
 4. To implement a Decision Tree Machine Learning model for difficulty prediction.
 5. To use selected features of DSA problems for Machine Learning-based classification.
 6. To provide progressive hints to assist students without immediately revealing the complete solution.
 7. To display example inputs and expected outputs for better understanding.
 8. To provide time and space complexity information for each problem.
 9. To develop a simple and user-friendly interface using Streamlit.
 10. To demonstrate the application of Machine Learning in an educational problem-solving platform.
-

1.4 Scope of the Project

The scope of AI DSA Tutor includes providing an interactive environment for practicing fundamental Data Structures and Algorithms.

The current system contains:

- 30 DSA problems

- 11 DSA topics
- 3 difficulty categories
- 5 Machine Learning features
- Decision Tree-based difficulty prediction
- Progressive hints
- Example input and output
- Time complexity
- Space complexity
- Interactive Streamlit interface

The system is mainly intended for students who are learning and practicing DSA.

Limitations of Scope

The current version does not provide:

- Automatic evaluation of user-written source code.
- Personalized learning based on a student's complete historical performance.
- A complete solution-generation system.
- A large-scale adaptive learning model.
- Real-time coding competition functionality.

These features can be considered for future development.

CHAPTER 2

Literature Survey

2.1 Existing DSA Learning Platforms

Several online platforms are available for learning and practicing Data Structures and Algorithms. Platforms such as online coding judges and algorithm visualization systems provide problem-solving exercises, explanations, and visual demonstrations.

VisuAlgo is an example of an interactive platform designed to help learners understand Data Structures and Algorithms through visualization and animation. It also provides training and quiz functionality for learners.

Such platforms demonstrate that interactive learning can make algorithmic concepts easier to explore. However, the proposed AI DSA Tutor focuses on combining problem practice with Machine Learning-based difficulty prediction and progressive hints in a compact learning interface.

2.2 Machine Learning for Classification

Machine Learning classification techniques are widely used to categorize data into predefined classes. A Decision Tree is a supervised learning algorithm that represents classification decisions through a tree-like structure.

The Decision Tree approach is suitable for the proposed project because the system needs to classify DSA problems into three categories:

- Easy
- Medium
- Hard

The model uses selected problem features as input and produces a predicted difficulty class.

Decision Trees also have the advantage of being relatively interpretable because their classification process can be represented through decision rules.

2.3 Interactive Learning Systems

Interactive learning systems allow students to actively engage with educational content rather than simply reading theoretical material.

For DSA learning, interactive systems can provide problem examples, visualizations, quizzes, hints, and immediate feedback. Such features can support self-paced learning and encourage students to attempt problems independently.

The AI DSA Tutor follows this approach by allowing students to select a problem, attempt it, and reveal hints progressively when required.

2.4 Research Gap

Existing DSA learning resources provide different types of functionality such as problem repositories, coding environments, tutorials, and algorithm visualizations.

However, the proposed system focuses on bringing multiple learning-support features together:

- DSA problem practice
- Difficulty prediction
- Progressive hints
- Complexity analysis
- Topic-based selection
- Interactive web interface

The main contribution of the project is the integration of a Machine Learning-based difficulty prediction component with a guided DSA practice environment.

CHAPTER 3

Proposed Methodology

3.1 System Design

The proposed AI DSA Tutor consists of several interconnected modules.

Overall Workflow

DSA Dataset



Data Preparation and Feature Selection



Decision Tree Model



Difficulty Prediction



Problem Selection



Problem Statement + Example



Student Attempts Problem



Progressive Hints



Complexity Analysis



Learning Support

The system is implemented as a Streamlit web application. Streamlit allows Python-based applications to provide interactive widgets and dynamically display content in a browser

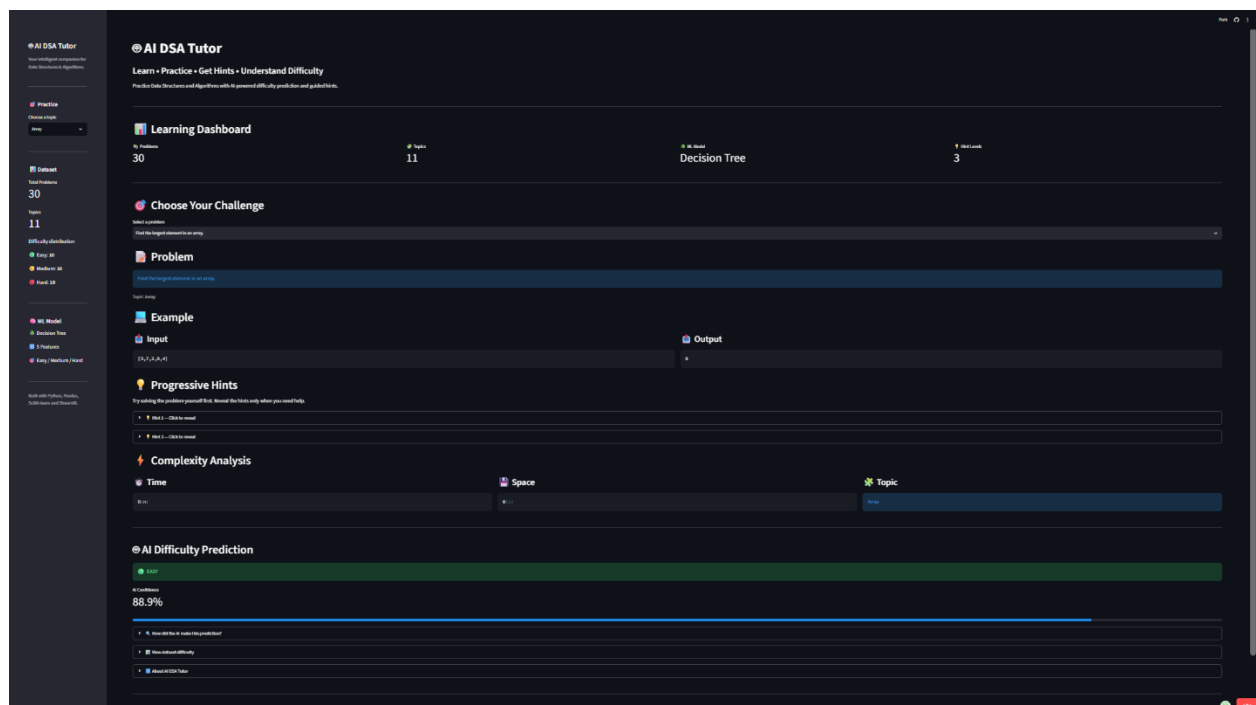


Figure 1: Overall Architecture of AI DSA Tutor

3.1.1 Problem Selection Module

The Problem Selection Module allows students to select a DSA topic from the available topics.

The system currently supports topics including:

- Array
- Binary Tree
- Dynamic Programming
- Graph
- Hashing
- Linked List
- Recursion
- Searching
- Sorting
- Stack
- Other related DSA topics

The selected topic is used to display an appropriate problem from the dataset.

3.1.2 Difficulty Prediction Module

The Difficulty Prediction Module is the Machine Learning component of the system.

A Decision Tree Classifier is trained using selected features associated with DSA problems.

The model predicts one of the three difficulty classes:

Easy → Medium → Hard

The application also displays an AI confidence score associated with the prediction.

For example, the current application interface displays a sample prediction of:

Predicted Difficulty: EASY

AI Confidence: 88.9%

This confidence value represents the model's confidence for that particular prediction and should not be interpreted as the overall accuracy of the Machine Learning model.

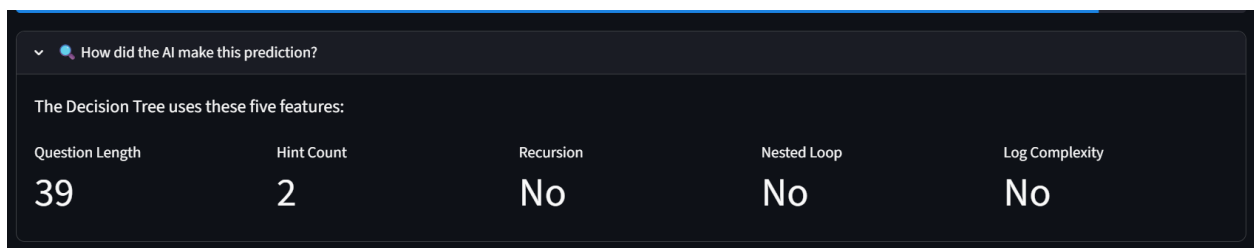


Figure 8: Decision Tree-based Prediction Workflow

3.1.3 Progressive Hint Module

The Progressive Hint Module is designed to help students without immediately exposing the complete answer.

The hints are arranged in levels.

A student can first attempt the problem independently. If the student is unable to proceed, the first hint can be revealed. If additional assistance is required, the next hint can be opened.

This approach encourages independent problem-solving while providing support when needed.

3.1.4 Complexity Analysis Module

The Complexity Analysis Module provides information about the efficiency of the expected solution.

It displays:

- Time Complexity
- Space Complexity

For example, the sample problem “Find the largest element in an array” is displayed with:

Time Complexity: $O(n)$

Space Complexity: $O(1)$

This helps students understand algorithm efficiency along with the problem-solving approach.

3.2 Modules Used

Module	Purpose
Dataset Module	Stores and manages DSA problem information
Machine Learning Module	Predicts problem difficulty
DSA Practice Module	Provides problems for practice
Progressive Hint Module	Provides step-by-step guidance
Complexity Analysis Module	Displays time and space complexity
User Interface Module	Provides the interactive Streamlit interface

Table 6: Project Modules

3.2.1 Dataset Module

The dataset contains DSA problem information required by the application.

The current dataset contains:

Table 1: Project Dataset Overview

Parameter	Value
Total Problems	30
Total Topics	11
Difficulty Levels	3
ML Features	5
Easy Problems	10
Medium Problems	10
Hard Problems	10

Sl. No. DSA Topic

1	Arrays
2	Binary Trees
3	Dynamic Programming
4	Graphs
5	Hashing
6	Linked Lists
7	Recursion
8	Searching
9	String

Sl. No.	DSA Topic
10	Stack
11	Tree

Table 2: DSA Topics Covered

Difficulty Level	Number of Problems
Easy	10
Medium	10
Hard	10
Total	30

Table 3: Difficulty Distribution

3.2.2 Machine Learning Module

The Machine Learning module uses a Decision Tree Classifier.

The basic process is:

1. Load the DSA dataset.
2. Select relevant features.
3. Prepare the target difficulty labels.
4. Train the Decision Tree model.
5. Provide problem features to the trained model.
6. Predict the difficulty level.
7. Display the prediction and confidence score.

3.2.3 DSA Practice Module

This module allows students to practice problems from different DSA topics.

Each selected problem provides:

- Problem title
 - Topic
 - Problem statement
 - Example input
 - Example output
 - Progressive hints
 - Complexity analysis
 - AI difficulty prediction
-

3.2.4 User Interface Module

The user interface is developed using Streamlit.

The interface contains a sidebar through which users can select a topic. The main page contains the learning dashboard, problem selection, problem details, hints, complexity analysis, and AI prediction sections.

3.3 Data Flow Diagram

A Data Flow Diagram represents how information moves between the user, dataset, Machine Learning model, and application modules.

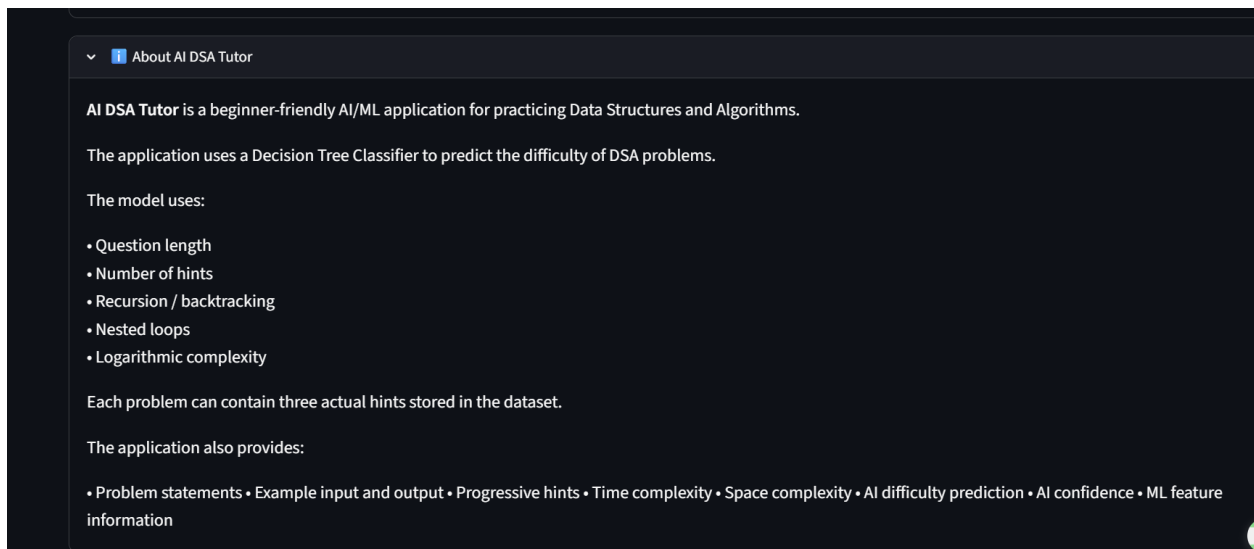


Figure 9: Overall Data Flow Diagram

3.3.1 DFD Level 0

External Entity

Student/User



AI DSA Tutor



DSA Problem Information + Hints + Complexity + Difficulty Prediction

The student interacts with the AI DSA Tutor through the web interface. The system retrieves the selected problem, processes the relevant features through the Machine Learning model, and displays learning information to the student.

3.3.2 DFD Level 1 – Problem Practice Module

Student



Select DSA Topic



Select Problem



Retrieve Problem Information



Display Problem + Example



Student Attempts Problem



Request Hint if Required



Display Progressive Hint

3.3.3 DFD Level 1 – Difficulty Prediction Module

DSA Problem Dataset



Feature Selection



Preprocessing



Decision Tree Model



Difficulty Prediction



Easy / Medium / Hard



Confidence Score



Display Prediction to Student

3.3.4 DFD Level 1 – Hint and Complexity Module

Selected Problem



Problem Information



Progressive Hints Complexity Analysis



Time Complexity + Space Complexity



Display Learning Support

3.4 Advantages

The major advantages of AI DSA Tutor are:

1. Interactive Learning: Students can practice DSA through an interactive web interface.
2. Topic-Based Practice: Problems can be selected according to DSA topics.
3. Difficulty Classification: Problems are categorized into Easy, Medium, and Hard.
4. AI-Based Prediction: The system uses a Decision Tree Machine Learning model to predict problem difficulty.
5. Progressive Assistance: Hints are revealed gradually instead of immediately providing a complete solution.
6. Complexity Awareness: Students can understand time and space complexity.
7. Self-Paced Learning: Students can attempt problems independently and seek help only when required.
8. Simple Interface: Streamlit provides a lightweight and accessible web interface.
9. Educational Application of ML: The project demonstrates a practical use of Machine Learning in education.

3.5 Requirement Specification

3.5.1 Hardware Requirements

Table 4: Hardware Requirements

Component Requirement

Processor	Intel i3 or equivalent and above
RAM	Minimum 4 GB
Storage	Minimum 10 GB free space
Display	Standard laptop/desktop display
Internet	Required for accessing the deployed application

3.5.2 Software Requirements

Software/Technology	Purpose
Python	Application and ML development
Pandas	Dataset handling and preprocessing
Scikit-learn	Machine Learning implementation
Decision Tree	Difficulty classification
Streamlit	Web application development
VS Code / Jupyter Notebook	Development environment
Web Browser	Accessing the application
Git/GitHub	Version control and project hosting

Table 5: Software Requirements

CHAPTER 4

Implementation and Result

4.1 Implementation of DSA Problem Selection

The application provides a topic selection option in the sidebar. Students can select a DSA category such as Array, Binary Tree, Dynamic Programming, Graph, Hashing, Linked List, Recursion, and other available topics.

After selecting a topic, the corresponding DSA problems are displayed through the problem selection interface.

The current application contains 30 problems across 11 topics.

Result

The topic selection functionality allows students to focus on a particular area of DSA rather than searching through unrelated problems.

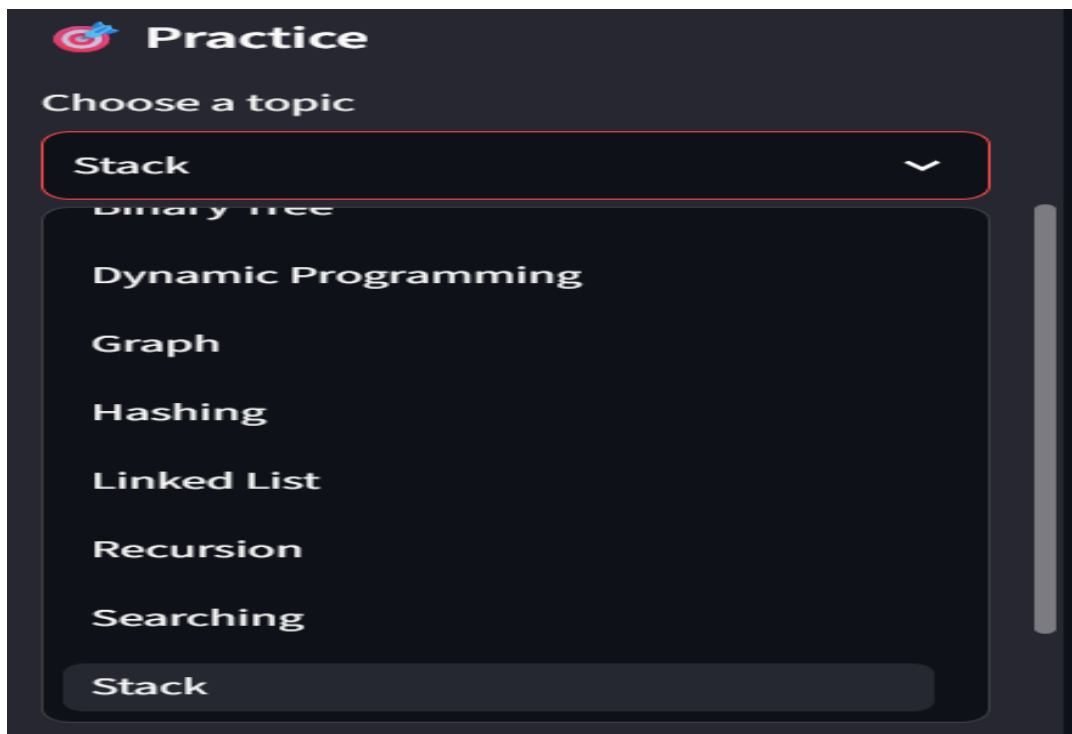


Figure 3: Topic and Problem Selection Interface

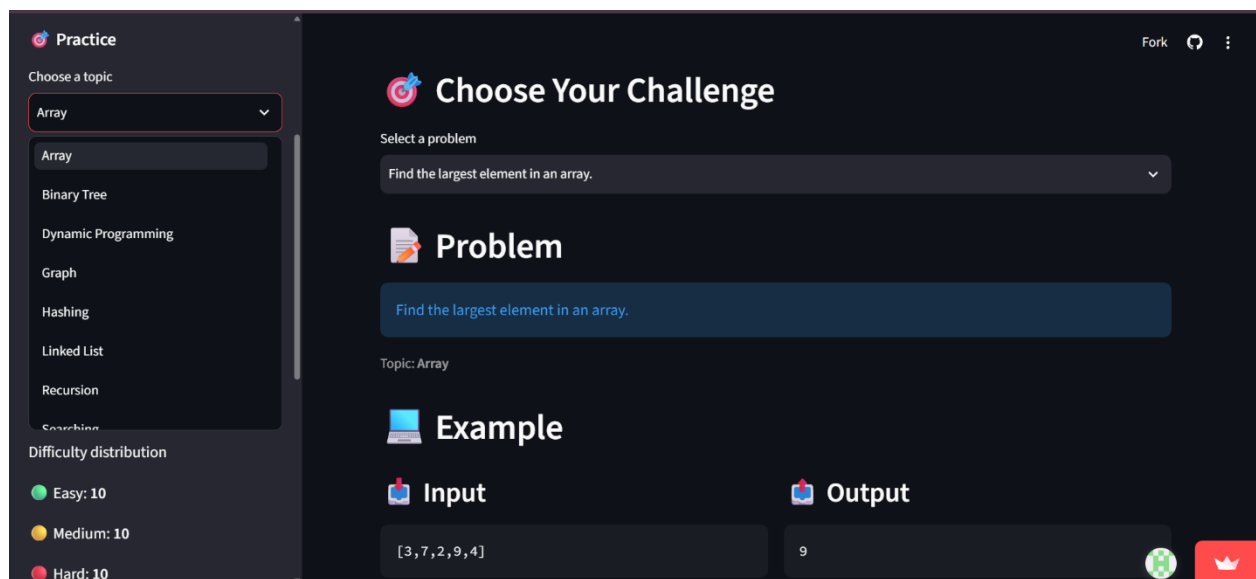


Figure 4: DSA Problem and Example Input/Output

4.2 Implementation of Difficulty Prediction

The Machine Learning component uses a Decision Tree model.

The selected problem is represented using five features. These features are provided to the trained model to predict the difficulty class.

The predicted class is displayed through the AI Difficulty Prediction section.

The three possible predictions are:

- Easy
- Medium
- Hard

The system also displays an AI confidence score.

Sample Result

For the problem “Find the largest element in an array”, the displayed result is:

Predicted Difficulty: EASY

AI Confidence: 88.9%

The result demonstrates that the Machine Learning model can provide a predicted difficulty category for a problem based on its selected features.

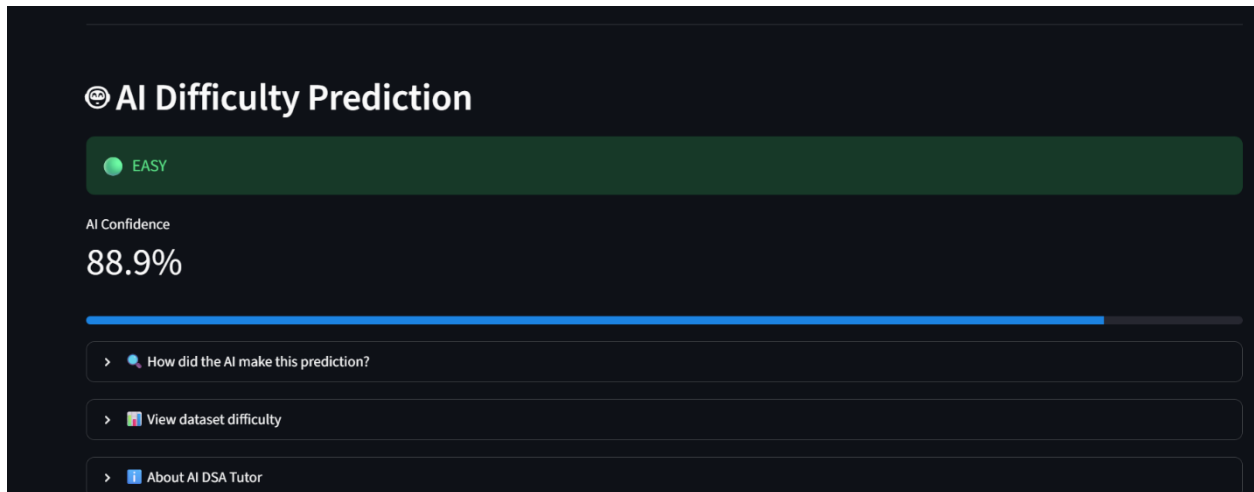


Figure 7: AI Difficulty Prediction Interface

Parameter		Result
Problem	Find the largest element in an array	
Predicted Difficulty	Easy	
AI Confidence	88.9%	
ML Model	Decision Tree	

Table 7: Sample Difficulty Prediction Result

4.3 Implementation of Progressive Hints

The system implements a progressive hint mechanism.

Initially, the hints remain hidden so that the student can attempt the problem independently.

When the student requires assistance, the hint sections can be expanded one by one.

This creates a learning process where the student receives the minimum necessary guidance before moving toward additional assistance.

Benefits

- Encourages independent thinking.
- Reduces immediate dependence on solutions.
- Helps students understand the approach step-by-step.
- Makes problem-solving more interactive.

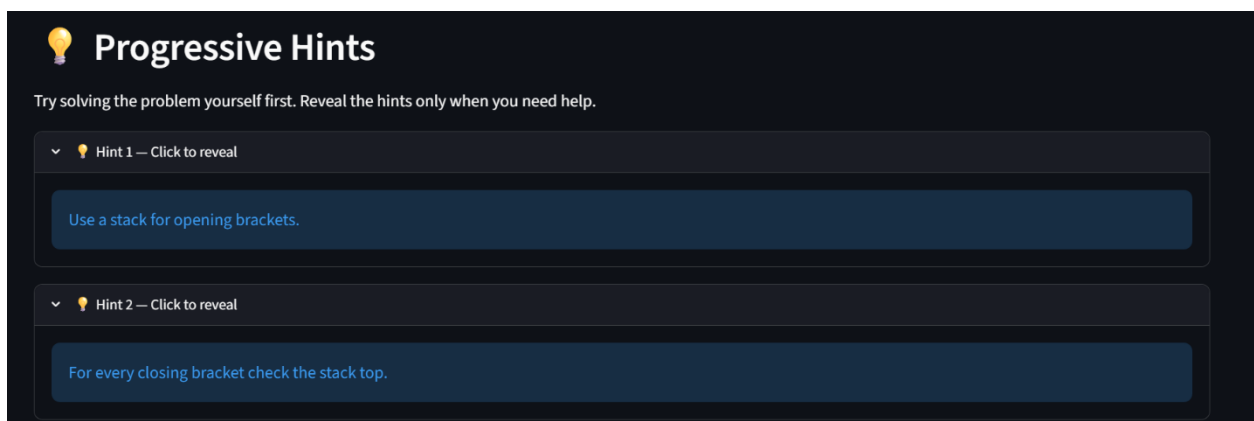


Figure 5: Progressive Hint Interface

4.4 Implementation of Complexity Analysis

The system provides complexity information for each problem.

The two main complexity measures are:

Time Complexity: Measures the approximate growth in computation required as the input size increases.

Space Complexity: Measures the additional memory required by the solution.

For the sample problem “Find the largest element in an array”, the application displays:

Complexity Value

Time $O(n)$

Space $O(1)$

This helps students develop awareness of algorithm efficiency.

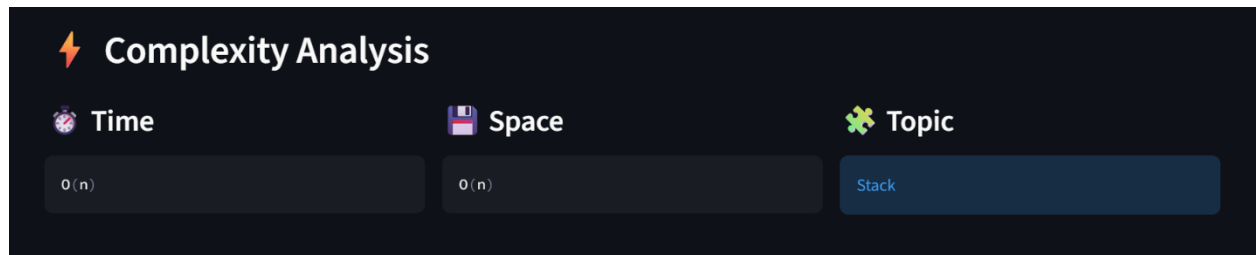


Figure 6: Complexity Analysis Interface

4.5 Web Application Interface

The web application contains a Learning Dashboard that provides an overview of the available learning content.

The dashboard displays:

- Number of problems
- Number of topics
- Machine Learning model
- Number of hint levels

The current dashboard displays:

Dashboard Component Result

Problems	30
Topics	11
ML Model	Decision Tree
Hint Levels	3

The sidebar also provides information about the dataset and difficulty distribution.

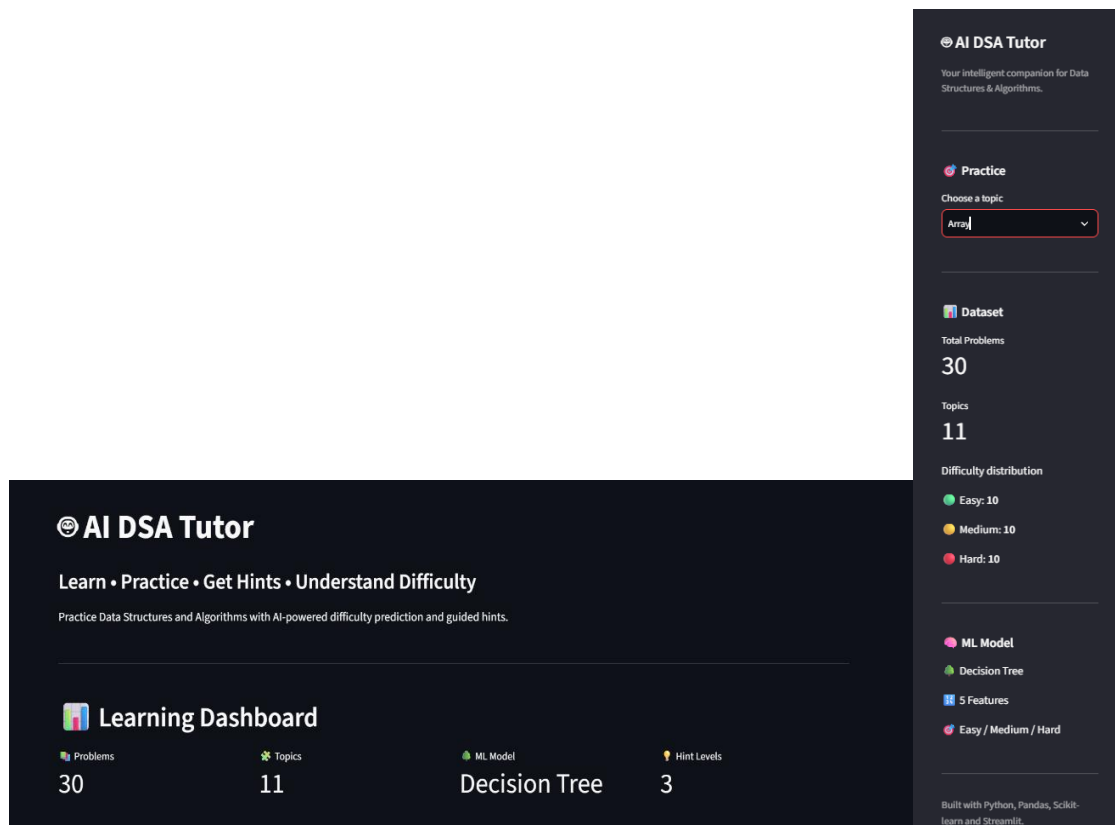


Figure 2: Learning Dashboard

4.6 Results and Observations

The developed prototype successfully integrates DSA practice with Machine Learning-based difficulty prediction.

The main observations are:

1. The application successfully provides 30 DSA problems.
2. The problems are distributed across 11 DSA topics.
3. The dataset contains 10 Easy, 10 Medium, and 10 Hard problems.
4. A Decision Tree is integrated as the Machine Learning model.
5. The system uses 5 features for its difficulty prediction component.
6. The application provides progressive hints to support students during problem solving.
7. Time and space complexity are displayed for the selected problems.
8. The application successfully provides an interactive interface through Streamlit.

9. The AI prediction interface displays both a difficulty category and a confidence score.

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

The development of AI DSA Tutor demonstrates that Machine Learning can be integrated with a DSA learning platform to provide additional learning support.

The major findings of the project are:

- A functional DSA learning platform was developed.
- Multiple DSA topics were integrated into one application.
- Problems were organized into three difficulty categories.
- A Decision Tree model was incorporated for difficulty prediction.
- Progressive hints were implemented to provide guided assistance.
- Complexity information was included to improve algorithmic understanding.
- Streamlit was successfully used to provide an interactive web interface.

The project therefore demonstrates the practical combination of DSA, Machine Learning, Python, and web application development.

5.2 GitHub Link and Website Link of the Project

GitHub Repository:

[<https://github.com/deeksha31005-spec/AI-DSA-Tutor>]

Website Link:

[<https://ai-dsa-tutor-xttcrtgy7kavje5n62mypq.streamlit.app/>]

5.3 Video Recording of Project Demonstration

Project Demonstration Video:

[<https://drive.google.com/file/d/1sYsA0Jn4nSNW1-ROZqgOplW8KgTrDZ27/view?usp=sharing>]

The demonstration video should show:

1. Opening the AI DSA Tutor.
2. Selecting a DSA topic.
3. Selecting a problem.
4. Viewing the example input and output.
5. Opening progressive hints.
6. Viewing time and space complexity.
7. Viewing the AI difficulty prediction and confidence score.

5.4 Limitations

Although the current system provides several useful features, it has certain limitations:

1. The current dataset contains only 30 problems, which limits the variety of available practice questions.
 2. The Machine Learning model is trained using a relatively small project-specific dataset.
 3. The current system predicts problem difficulty based on predefined features and does not continuously learn from individual student performance.
 4. The application does not currently evaluate a student's submitted source code automatically.
 5. The system does not provide complete personalized learning paths based on a student's previous performance.
 6. The quality of difficulty prediction depends on the quality and representation of the dataset and selected features.
 7. The current hint system provides predefined guidance rather than dynamically generating hints based on a student's actual code or attempt.
-

5.5 Future Work

The project can be enhanced in several ways in the future.

1. Larger Dataset

The number of DSA problems can be increased significantly to provide more practice opportunities and improve the Machine Learning model.

2. Personalized Learning

The system can maintain a student's performance history and recommend problems based on strengths and weaknesses.

3. Automatic Code Evaluation

A coding environment can be added where students submit their solutions and receive feedback regarding correctness and complexity.

4. AI-Generated Hints

Instead of using only predefined hints, Generative AI can be integrated to provide hints based on the student's approach without directly revealing the complete solution.

5. Adaptive Difficulty

The system can automatically increase or decrease problem difficulty according to the student's performance.

6. Performance Dashboard

A student dashboard can be developed to display:

- Problems attempted
- Problems solved

- Success rate
- Topic-wise performance
- Difficulty-wise performance
- Progress over time

7. Improved Machine Learning Models

Other classification algorithms can be evaluated and compared with the Decision Tree model to determine which approach performs better on a larger dataset.

8. Gamification

Points, badges, streaks, levels, and achievement systems can be added to make DSA practice more engaging.

5.6 Conclusion

The AI DSA Tutor project presents an interactive approach to Data Structures and Algorithms learning by combining problem practice, progressive guidance, complexity analysis, and Machine Learning-based difficulty prediction.

The system successfully provides a structured collection of DSA problems across multiple topics and difficulty levels. The Decision Tree model adds a Machine Learning component by predicting the difficulty of problems using selected features. The progressive hint mechanism encourages students to attempt problems independently while still providing assistance when necessary.

The use of Streamlit provides a simple and accessible web interface, while Python, Pandas, and Scikit-learn support the data processing and Machine Learning components.

Overall, the project demonstrates how Machine Learning can be applied to an educational platform to improve the organization and delivery of DSA practice. With future improvements such as personalized recommendations, automatic code evaluation, AI-generated hints, adaptive difficulty, and a larger dataset, the system can be developed into a more comprehensive intelligent DSA learning platform.

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APPENDICES

Appendix A – Sample DSA Problem

Problem: Find the largest element in an array.

Input:

[3, 7, 2, 9, 4]

Expected Output:

9

Topic: Array

Time Complexity: $O(n)$

Space Complexity: $O(1)$

Appendix B – Sample Difficulty Prediction

Parameter	Result
Problem	Find the largest element in an array
Predicted Difficulty	Easy
AI Confidence	88.9%
ML Model	Decision Tree

Appendix C – Dataset Overview

Category	Count
Total Problems	30
Total Topics	11

Easy	10
Medium	10
Hard	10
ML Features	5

Appendix D – Project Development Timeline

Day	Work Completed
Day 1	Problem identification, requirement analysis, and project planning
Day 2	DSA dataset preparation and feature selection
Day 3	Data preprocessing and Decision Tree model development
Day 4	DSA problem practice and topic selection module
Day 5	Progressive hints and complexity analysis
Day 6	Streamlit integration, UI development, testing, and debugging
Day 7	Final testing, improvements, deployment, and documentation

Table 8: Project Development Timeline